Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
F40	Q_{desc}	Root	9.6	6.7x	6.9x	3.0	1.6x	1.6x	2.0	0.99x	0.88x
F40	Q_{desc}	H-LP	8.8	6.3x	6.4x	2.7	1.9x	2.1x	1.3	0.86x	0.87x
F40	Q_{desc}	MLP	8.9	6.4x	6.4x	2.7	1.4x	1.6x	1.3	0.86x	0.78x
F1K	Q_{desc}	Root	440	190x	190x	6.3	2.1x	2.5x	30	1.0x	0.71x
F1K	Q_{desc}	H-LP	13	5.9x	5.9x	4.2	2.2x	1.9x	1.9	0.68x	0.85x
F1K	Q_{desc}	MLP	13	6.7x	7.1x	3.6	1.7x	1.8x	1.8	1.0x	0.95x
NT10	Q_{desc}	Root	9.3	7.0x	6.9x	2.8	1.4x	1.6x	2.2	1.0x	0.96x
NT10	Q_{desc}	H/L	9.3	7.0x	6.9x	2.6	1.5x	1.7x	1.4	0.83x	0.88x
NT100	Q_{desc}	Root	20	13x	13x	3.9	2.2x	2.2x	11	0.86x	0.84x
NT100	Q_{desc}	H-LP	9.9	6.8x	7.0x	3.1	1.7x	1.9x	2.2	0.91x	0.94x
NT100	Q_{desc}	MLP	8.8	6.1x	6.2x	2.9	1.9x	1.8x	1.5	0.95x	0.83x
NT1K	Q_{desc}	Root	1.1K	400x	390x	10	5.2x	4.9x	58	1.3x	1.2x
NT1K	Q_{desc}	H-LP	16	6.7x	6.8x	2.9	1.9x	1.7x	1.3	0.93x	0.90x
NT1K	Q_{desc}	MLP	12	5.4x	5.5x	3.8	2.2x	2.1x	1.2	0.99x	0.90x
NT10K	Q_{desc}	Root	110K	6.9Kx	6.9Kx	100	9.5x	11x	420	0.85x	0.78x
NT10K	Q_{desc}	H-LP	270	23x	28x	2.7	0.56x	1.1x	1.4	1.0x	1.0x
NT10K	Q_{desc}	MLP	250	21x	25x	3.0	0.51x	1.4x	1.2	0.67x	0.65x
NT100K	Q_{desc}	Root	19M	130Kx	140Kx	260	2.8x	3.4x	4.3K	0.82x	0.81x
NT100K	Q_{desc}	H-LP	2.5K	18x	20x	3.8	0.07x	0.17x	2.1	0.86x	0.65x
NT100K	Q_{desc}	MLP	470	3.4x	4.1x	2.5	0.05x	0.09x	1.4	0.90x	0.77x
DT10	Q_{desc}	Root	9.5	7.0x	6.7x	2.9	2.0x	1.7x	1.2	0.54x	0.53x
DT10	Q_{desc}	H-LP	9.3	6.8x	6.8x	2.8	1.8x	1.8x	1.7	0.91x	0.89x
DT10	Q_{desc}	MLP	9.0	6.6x	6.7x	2.8	1.8x	1.7x	1.3	0.97x	0.97x
DT100	Q_{desc}	Root	20	14x	14x	3.9	2.6x	2.2x	12	0.86x	0.84x
DT100	Q_{desc}	H-LP	9.3	6.5x	6.6x	3.0	2.0x	1.8x	1.7	0.95x	0.95x
DT100	Q_{desc}	MLP	9.3	6.5x	6.4x	2.8	1.8x	1.7x	1.5	0.94x	0.92x
DT1K	Q_{desc}	Root	1.1K	390x	390x	11	6.6x	6.2x	40	0.84x	0.81x
DT1K	Q_{desc}	H-LP	14	5.9x	6.2x	3.5	2.1x	2.2x	0.86	0.95x	0.99x
DT1K	Q_{desc}	MLP	11	4.7x	4.9x	3.7	2.4x	2.4x	0.72	0.92x	0.69x
DT10K	Q_{desc}	Root	110K	6.7Kx	7.0Kx	32	4.5x	3.6x	420	0.85x	0.82x
DT10K	Q_{desc}	H-LP	250	21x	25x	2.6	0.56x	0.98x	1.2	0.89x	0.77x
DT10K	Q_{desc}	MLP	220	19x	24x	2.6	0.49x	0.92x	1.4	0.93x	1.2x
DT100K	Q_{desc}	Root	**>6 h**	**>150Kx**	**>150Kx**	130	1.5x	1.7x	4.3K	0.83x	0.80x
DT100K	Q_{desc}	H-LP	800	5.6x	6.6x	3.4	0.07x	0.13x	1.3	0.74x	0.76x
DT100K	Q_{desc}	MLP	470	3.4x	3.9x	2.3	0.04x	0.17x	1.6	0.99x	1.0x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
WT10	Q_{desc}	Root	9.4	6.8x	6.6x	3.0	1.9x	2.0x	2.0	0.93x	0.91x
WT10	Q_{desc}	H-LP	9.3	6.6x	7.0x	3.1	1.9x	2.1x	1.7	0.89x	0.85x
WT10	Q_{desc}	MLP	9.0	6.6x	6.6x	2.8	2.0x	1.9x	1.4	0.97x	0.96x
WT100	Q_{desc}	Root	26	12x	14x	3.5	2.3x	2.3x	12	0.88x	0.86x
WT100	Q_{desc}	H-LP	10	7.1x	7.1x	2.7	1.6x	1.9x	2.5	0.93x	0.91x
WT100	Q_{desc}	MLP	12	6.7x	6.8x	3.0	2.0x	2.0x	1.7	0.82x	0.95x
WT1K	Q_{desc}	Root	1.2K	440x	420x	8.7	5.3x	3.9x	59	1.3x	1.3x
WT1K	Q_{desc}	H-LP	20	8.8x	8.7x	3.3	1.7x	1.8x	1.6	0.77x	0.60x
WT1K	Q_{desc}	MLP	22	9.7x	13x	3.5	2.2x	2.0x	2.2	0.93x	0.88x
WT10K	Q_{desc}	Root	110K	13Kx	10Kx	25	2.4x	2.9x	410	0.84x	0.81x
WT10K	Q_{desc}	H-LP	330	29x	33x	2.4	0.47x	0.81x	1.7	0.78x	0.61x
WT10K	Q_{desc}	MLP	110	20x	60x	3.0	0.65x	1.0x	2.4	0.92x	0.93x
WT100K	Q_{desc}	Root	24M	310Kx	320Kx	120	1.3x	1.6x	4.3K	0.84x	0.82x
WT100K	Q_{desc}	H-LP	3.0K	21x	23x	3.5	0.07x	0.13x	2.3	0.89x	0.90x
WT100K	Q_{desc}	MLP	580	15x	300x	2.2	0.04x	0.08x	1.2	0.93x	0.92x
SNB/C	Q_{desc}	Root	92K	140x	47Kx	3.6	0.01x	0.01x	4.7	1.0x	1.0x
SNB/C	Q_{desc}	D-r.	82K	31x	39x	4.1	0.01x	0.01x	4.1	0.92x	1.0x
SNB/C	Q_{desc}	H-LP	69K	26x	28x	3.6	0.01x	0.01x	3.5	0.93x	0.87x
SNB/C	Q_{desc}	MLP	22K	31x	11Kx	3.4	0.01x	0.01x	1.6	1.0x	0.99x
SNB/P	Q_{desc}	Root	17K	6.2Kx	6.6Kx	6.9	2.8x	3.1x	46	1.1x	1.1x
SNB/P	Q_{desc}	H-LP	15K	5.1Kx	5.1Kx	3.7	1.9x	2.0x	11	0.85x	0.79x
SNB/P	Q_{desc}	MLP	15K	9.0Kx	7.8Kx	3.8	1.8x	2.0x	1.1	0.87x	0.74x
SNB/T	Q_{desc}	Root	14K	7.7Kx	7.6Kx	3.3	2.0x	2.1x	9.7	0.89x	0.85x
SNB/T	Q_{desc}	H-LP	15K	9.9Kx	10Kx	3.1	1.9x	1.8x	8.6	0.96x	0.94x
SNB/T	Q_{desc}	MLP	14K	9.0Kx	8.6Kx	2.9	1.9x	2.0x	1.5	1.0x	1.0x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
F40	Q_{child}	High	2.0	1.2x	1.3x	1.7	0.91x	0.93x	1.7	1.1x	1.0x
F40	Q_{child}	Low	1.9	1.2x	1.3x	2.0	1.2x	1.0x	1.2	1.1x	1.1x
F1K	Q_{child}	High	2.4	0.97x	1.0x	1.6	0.63x	0.64x	1.9	0.99x	1.2x
F1K	Q_{child}	Low	2.3	0.95x	1.0x	1.5	0.65x	1.1x	1.1	0.97x	0.96x
NT10	Q_{child}	High	1.9	1.2x	1.5x	1.7	0.82x	0.89x	1.6	0.94x	0.94x
NT10	Q_{child}	Low	1.9	1.2x	1.3x	1.6	0.94x	1.3x	1.3	1.2x	1.2x
NT100	Q_{child}	High	2.0	1.3x	1.3x	1.8	1.1x	1.1x	2.0	0.97x	0.91x
NT100	Q_{child}	Low	1.9	1.2x	1.3x	1.7	1.0x	0.95x	1.3	1.1x	0.89x
NT1K	Q_{child}	High	2.4	1.0x	1.0x	1.7	0.89x	0.88x	1.2	1.1x	1.1x
NT1K	Q_{child}	Low	2.2	0.97x	1.0x	1.7	0.91x	0.96x	0.71	0.97x	1.0x
NT10K	Q_{child}	High	6.3	0.60x	0.61x	2.2	0.49x	0.72x	1.9	0.98x	1.0x
NT10K	Q_{child}	Low	5.4	0.52x	0.53x	2.2	0.43x	0.91x	1.1	0.94x	0.97x
NT100K	Q_{child}	High	57	0.53x	0.55x	2.2	0.06x	0.08x	3.5	0.92x	0.86x
NT100K	Q_{child}	Low	56	0.53x	0.54x	1.9	0.05x	0.07x	1.3	0.93x	0.98x
DT10	Q_{child}	High	2.0	1.3x	1.3x	1.8	1.3x	1.2x	1.5	0.89x	0.95x
DT10	Q_{child}	Low	1.9	1.2x	1.3x	1.6	0.90x	0.92x	1.1	0.97x	0.97x
DT100	Q_{child}	High	2.0	1.2x	1.3x	1.4	0.99x	0.82x	1.7	1.1x	1.1x
DT100	Q_{child}	Low	1.9	1.2x	1.3x	1.9	1.2x	1.1x	1.0	0.94x	0.94x
DT1K	Q_{child}	High	2.4	1.0x	1.1x	1.1	0.64x	0.60x	1.5	0.98x	1.1x
DT1K	Q_{child}	Low	2.3	0.94x	1.0x	1.8	1.0x	1.0x	0.95	0.93x	0.92x
DT10K	Q_{child}	High	6.3	0.60x	0.61x	1.7	0.34x	0.59x	1.5	1.0x	1.0x
DT10K	Q_{child}	Low	5.4	0.52x	0.53x	1.8	0.41x	0.67x	1.0	1.0x	1.0x
DT100K	Q_{child}	High	65	0.62x	0.66x	1.7	0.04x	0.08x	1.8	1.0x	1.1x
DT100K	Q_{child}	Low	56	0.53x	0.54x	1.8	0.05x	0.09x	1.2	0.90x	1.0x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
WT10	Q_{child}	High	2.0	1.3x	1.4x	2.2	1.2x	1.3x	1.7	0.97x	0.99x
WT10	Q_{child}	Low	1.8	1.2x	1.3x	1.7	1.0x	1.3x	1.1	0.99x	0.95x
WT100	Q_{child}	High	2.0	1.2x	1.3x	1.6	0.92x	0.87x	2.0	0.91x	1.1x
WT100	Q_{child}	Low	2.0	1.2x	1.3x	1.6	1.1x	1.1x	1.1	0.98x	0.96x
WT1K	Q_{child}	High	2.5	1.0x	1.0x	1.6	0.89x	0.86x	2.3	0.95x	0.99x
WT1K	Q_{child}	Low	2.3	1.0x	0.99x	1.6	0.93x	1.0x	1.1	0.95x	0.92x
WT10K	Q_{child}	High	6.3	0.59x	0.60x	1.5	0.29x	0.51x	2.6	0.88x	0.96x
WT10K	Q_{child}	Low	5.4	0.52x	0.52x	1.5	0.35x	0.58x	1.2	0.98x	0.98x
WT100K	Q_{child}	High	63	0.62x	0.63x	1.6	0.04x	0.06x	1.5	0.87x	0.84x
WT100K	Q_{child}	Low	54	0.52x	0.55x	2.0	0.05x	0.09x	0.74	1.1x	1.1x
SNB/C	Q_{child}	High	1.4K	0.40x	0.41x	2.8	0.01x	0.01x	3.4	0.90x	0.89x
SNB/C	Q_{child}	Low	1.2K	0.33x	0.36x	1.8	0.01x	0.01x	1.2	0.92x	1.2x
SNB/P	Q_{child}	High	3.5	1.1x	1.2x	2.4	1.0x	1.1x	24	0.95x	0.93x
SNB/P	Q_{child}	Low	2.4	0.82x	0.94x	1.5	0.62x	0.63x	1.2	0.91x	0.92x
SNB/T	Q_{child}	High	2.1	1.3x	1.4x	1.7	1.0x	0.94x	3.7	0.89x	0.90x
SNB/T	Q_{child}	Low	1.9	1.2x	1.3x	1.7	1.1x	1.0x	1.2	1.1x	0.94x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
F40	Q_{leaf}	Root	9.6	6.1x	4.7x	5.3	1.7x	3.2x	1.9	1.0x	1.1x
F40	Q_{leaf}	H-LP	9.9	6.5x	4.8x	4.6	1.7x	2.3x	1.4	0.94x	0.94x
F40	Q_{leaf}	MLP	9.6	6.5x	4.7x	4.2	1.3x	2.2x	1.4	0.96x	0.84x
F1K	Q_{leaf}	Root	310	16x	130x	8.0	2.7x	3.6x	11	0.43x	0.43x
F1K	Q_{leaf}	H-LP	96	40x	43x	5.6	1.6x	2.9x	2.3	0.86x	1.1x
F1K	Q_{leaf}	MLP	95	40x	42x	6.4	2.4x	3.6x	1.9	0.96x	1.1x
NT10	Q_{leaf}	Root	9.7	6.6x	6.9x	3.8	1.4x	2.0x	1.8	0.96x	0.98x
NT10	Q_{leaf}	H/L	9.0	6.3x	6.3x	4.6	1.6x	2.4x	1.6	0.93x	1.1x
NT100	Q_{leaf}	Root	18	6.4x	2.9x	6.1	2.0x	3.8x	9.4	0.84x	0.96x
NT100	Q_{leaf}	H-LP	11	7.4x	1.8x	5.0	1.7x	2.8x	2.2	0.91x	1.1x
NT100	Q_{leaf}	MLP	11	7.2x	1.8x	5.4	1.8x	3.3x	1.8	1.0x	1.1x
NT1K	Q_{leaf}	Root	950	7.5x	340x	13	3.9x	5.6x	31	0.88x	0.87x
NT1K	Q_{leaf}	H-LP	160	71x	70x	4.5	1.2x	2.2x	1.4	0.86x	0.96x
NT1K	Q_{leaf}	MLP	160	65x	69x	5.2	1.5x	3.0x	1.2	1.0x	1.1x
NT10K	Q_{leaf}	Root	93K	7.4x	6.4Kx	49	2.4x	6.4x	290	0.87x	0.84x
NT10K	Q_{leaf}	H-LP	12K	1.0Kx	1.1Kx	5.7	0.32x	2.3x	1.4	0.75x	0.66x
NT10K	Q_{leaf}	MLP	12K	1.0Kx	1.1Kx	5.1	0.32x	2.2x	1.8	0.92x	0.96x
NT100K	Q_{leaf}	Root	19M	16x	170Kx	110	0.53x	1.5x	4.1K	0.83x	0.81x
NT100K	Q_{leaf}	H-LP	1.2M	3.6Kx	11Kx	5.7	0.04x	0.27x	2.4	0.97x	1.0x
NT100K	Q_{leaf}	MLP	1.2M	3.6Kx	11Kx	4.6	0.03x	0.23x	1.3	1.0x	1.2x
DT10	Q_{leaf}	Root	9.4	6.2x	6.5x	3.9	1.5x	2.3x	1.8	0.97x	0.90x
DT10	Q_{leaf}	H-LP	9.6	6.8x	6.7x	3.6	1.4x	2.2x	2.0	1.0x	1.0x
DT10	Q_{leaf}	MLP	9.7	7.4x	6.6x	3.9	1.5x	2.3x	1.5	0.95x	1.0x
DT100	Q_{leaf}	Root	16	5.7x	3.0x	4.5	1.6x	2.5x	6.8	0.75x	0.83x
DT100	Q_{leaf}	H-LP	11	7.3x	2.1x	4.1	1.6x	2.5x	1.8	0.94x	1.0x
DT100	Q_{leaf}	MLP	11	7.1x	2.0x	4.1	1.4x	2.3x	1.6	0.96x	1.0x
DT1K	Q_{leaf}	Root	650	5.4x	240x	13	4.7x	6.5x	38	1.1x	1.3x
DT1K	Q_{leaf}	H-LP	97	40x	43x	4.8	1.5x	2.4x	1.3	0.72x	0.72x
DT1K	Q_{leaf}	MLP	96	41x	42x	5.1	1.8x	2.8x	1.6	0.99x	1.0x
DT10K	Q_{leaf}	Root	60K	5.5x	4.5Kx	30	2.9x	5.7x	200	0.87x	0.86x
DT10K	Q_{leaf}	H-LP	6.4K	550x	610x	3.8	0.29x	2.1x	1.3	0.81x	0.70x
DT10K	Q_{leaf}	MLP	6.4K	550x	610x	5.2	0.29x	2.7x	1.6	1.0x	1.0x
DT100K	Q_{leaf}	Root	12M	13x	120Kx	140	0.87x	3.2x	2.1K	0.83x	0.81x
DT100K	Q_{leaf}	H-LP	620K	1.8Kx	5.6Kx	5.9	0.05x	0.51x	1.3	0.88x	0.98x
DT100K	Q_{leaf}	MLP	620K	1.8Kx	5.7Kx	4.5	0.04x	0.36x	1.1	0.94x	1.1x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
WT10	Q_{leaf}	Root	9.6	6.4x	6.5x	4.1	1.5x	2.6x	1.9	0.96x	1.0x
WT10	Q_{leaf}	H-LP	9.2	6.5x	6.4x	4.3	1.5x	2.3x	1.6	0.88x	0.90x
WT10	Q_{leaf}	MLP	9.6	6.7x	6.7x	3.7	1.5x	2.5x	1.5	0.94x	0.99x
WT100	Q_{leaf}	Root	20	7.1x	2.7x	5.0	1.7x	2.7x	7.3	0.65x	0.61x
WT100	Q_{leaf}	H-LP	12	7.6x	1.6x	3.7	1.4x	1.9x	2.8	0.91x	0.94x
WT100	Q_{leaf}	MLP	11	7.1x	1.5x	3.8	1.4x	2.0x	2.1	0.94x	0.98x
WT1K	Q_{leaf}	Root	1.1K	8.4x	410x	9.4	3.0x	4.9x	45	1.1x	1.1x
WT1K	Q_{leaf}	H-LP	170	70x	73x	4.7	1.4x	2.4x	1.7	0.94x	0.98x
WT1K	Q_{leaf}	MLP	170	69x	72x	4.8	1.4x	3.0x	1.5	0.94x	0.98x
WT10K	Q_{leaf}	Root	110K	8.1x	7.2Kx	30	1.2x	3.7x	380	0.94x	0.82x
WT10K	Q_{leaf}	H-LP	13K	1.1Kx	1.1Kx	3.7	0.18x	1.5x	2.1	0.78x	0.64x
WT10K	Q_{leaf}	MLP	13K	1.1Kx	1.1Kx	4.5	0.21x	1.8x	2.8	1.0x	1.3x
WT100K	Q_{leaf}	Root	18M	17x	160Kx	110	0.58x	1.6x	4.0K	0.84x	0.81x
WT100K	Q_{leaf}	H-LP	1.2M	3.6Kx	12Kx	5.4	0.03x	0.21x	2.3	0.78x	0.54x
WT100K	Q_{leaf}	MLP	1.2M	3.6Kx	11Kx	4.3	0.03x	0.21x	1.8	0.90x	0.94x
SNB/C	Q_{leaf}	Root	**>6 h**	**>8.0Kx**	**>9.2Kx**	6.0	0.01x	0.02x	2.0	0.66x	0.70x
SNB/C	Q_{leaf}	D-r.	**>6 h**	**>7.9Kx**	**>9.2Kx**	5.9	0.01x	0.02x	2.7	0.91x	0.86x
SNB/C	Q_{leaf}	H-LP	**>6 h**	**>8.0Kx**	**>9.2Kx**	5.2	0.01x	0.01x	3.6	0.93x	0.93x
SNB/C	Q_{leaf}	MLP	**>6 h**	**>7.9Kx**	**>9.2Kx**	5.9	0.01x	0.02x	1.6	0.98x	1.0x
SNB/P	Q_{leaf}	Root	17K	210x	5.3Kx	9.6	2.2x	3.8x	46	1.1x	1.1x
SNB/P	Q_{leaf}	H-LP	15K	1.8Kx	5.2Kx	5.1	1.3x	2.4x	12	0.81x	0.76x
SNB/P	Q_{leaf}	MLP	15K	5.1Kx	5.3Kx	5.0	1.2x	2.2x	1.2	0.80x	0.80x
SNB/T	Q_{leaf}	Root	15K	6.4Kx	3.3Kx	4.6	1.9x	2.6x	10	1.0x	1.1x
SNB/T	Q_{leaf}	H-LP	15K	7.3Kx	3.4Kx	4.4	1.7x	2.6x	8.2	1.3x	0.95x
SNB/T	Q_{leaf}	MLP	14K	9.5Kx	3.4Kx	3.4	1.4x	2.0x	1.7	1.1x	1.1x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
F40	$Q_{a\&d}$	RL	10	7.3x	7.9x	22	25x	24x	1.5	1.0x	1.1x
F40	$Q_{a\&d}$	DPL	10	9.2x	8.1x	20	22x	21x	1.3	0.98x	0.87x
F40	$Q_{a\&d}$	Sib.	9.9	9.0x	9.1x	20	21x	22x	1.5	1.2x	1.1x
F40	$Q_{a\&d}$	Dist	9.3	6.7x	8.5x	21	24x	23x	1.5	1.1x	1.0x
F1K	$Q_{a\&d}$	RL	12	8.8x	8.4x	19	19x	21x	1.7	1.5x	1.1x
F1K	$Q_{a\&d}$	DPL	12	7.5x	8.4x	22	22x	24x	1.8	1.6x	1.2x
F1K	$Q_{a\&d}$	Sib.	11	4.2x	4.2x	17	16x	19x	1.6	1.2x	0.96x
F1K	$Q_{a\&d}$	Dist	10	4.1x	4.0x	22	23x	21x	1.7	1.3x	1.0x
NT10	$Q_{a\&d}$	RL	9.8	8.0x	9.1x	19	20x	20x	1.6	1.2x	1.3x
NT10	$Q_{a\&d}$	DPL	9.8	8.9x	7.9x	19	19x	20x	1.3	1.1x	1.2x
NT10	$Q_{a\&d}$	Sib.	9.3	7.3x	9.5x	21	21x	21x	1.5	1.0x	0.87x
NT10	$Q_{a\&d}$	Dist	9.4	9.4x	9.4x	20	23x	22x	1.4	1.2x	1.1x
NT100	$Q_{a\&d}$	RL	10	7.6x	7.6x	21	22x	23x	2.0	1.5x	2.7x
NT100	$Q_{a\&d}$	DPL	10	7.9x	7.7x	21	22x	21x	1.4	1.1x	0.96x
NT100	$Q_{a\&d}$	Sib.	9.3	8.0x	8.2x	19	20x	20x	1.7	1.3x	1.5x
NT100	$Q_{a\&d}$	Dist	9.5	8.3x	8.5x	22	24x	23x	1.2	1.3x	1.3x
NT1K	$Q_{a\&d}$	RL	11	7.4x	7.9x	20	20x	22x	1.2	1.5x	1.7x
NT1K	$Q_{a\&d}$	DPL	11	7.6x	8.1x	19	23x	22x	0.86	0.78x	0.79x
NT1K	$Q_{a\&d}$	Sib.	11	4.2x	3.9x	22	24x	26x	1.4	1.3x	1.2x
NT1K	$Q_{a\&d}$	Dist	11	4.2x	4.1x	21	23x	22x	1.4	1.4x	1.5x
NT10K	$Q_{a\&d}$	RL	18	12x	13x	22	26x	24x	1.6	1.2x	1.2x
NT10K	$Q_{a\&d}$	DPL	18	12x	11x	22	26x	26x	1.3	1.2x	1.2x
NT10K	$Q_{a\&d}$	Sib.	21	1.3x	1.3x	21	24x	23x	1.4	1.1x	1.1x
NT10K	$Q_{a\&d}$	Dist	21	1.3x	1.3x	25	30x	26x	1.6	1.3x	1.3x
NT100K	$Q_{a\&d}$	RL	79	49x	48x	20	18x	23x	1.1	1.2x	1.4x
NT100K	$Q_{a\&d}$	DPL	79	49x	49x	22	23x	24x	0.93	1.2x	1.1x
NT100K	$Q_{a\&d}$	Sib.	130	0.80x	0.80x	18	18x	21x	0.84	1.0x	1.0x
NT100K	$Q_{a\&d}$	Dist	130	0.79x	0.79x	20	19x	21x	0.90	1.1x	1.1x
DT10	$Q_{a\&d}$	RL	9.8	7.3x	7.4x	17	20x	18x	1.4	1.2x	1.3x
DT10	$Q_{a\&d}$	DPL	9.7	9.0x	7.6x	18	21x	18x	1.3	1.1x	1.1x
DT10	$Q_{a\&d}$	Sib.	9.4	8.8x	9.2x	19	22x	20x	1.2	1.0x	1.0x
DT10	$Q_{a\&d}$	Dist	9.4	9.2x	9.6x	18	21x	20x	1.3	1.1x	1.1x
DT100	$Q_{a\&d}$	RL	9.7	8.5x	7.3x	20	24x	23x	1.5	1.3x	1.4x
DT100	$Q_{a\&d}$	DPL	10	7.5x	7.6x	22	23x	23x	1.2	1.1x	1.0x
DT100	$Q_{a\&d}$	Sib.	9.5	8.0x	8.3x	19	19x	22x	1.3	1.1x	1.0x
DT100	$Q_{a\&d}$	Dist	9.6	8.1x	8.5x	19	20x	20x	1.4	1.1x	2.1x
DT1K	$Q_{a\&d}$	RL	11	7.6x	8.6x	20	24x	21x	1.8	1.3x	1.5x
DT1K	$Q_{a\&d}$	DPL	11	8.7x	9.0x	20	21x	21x	1.3	0.97x	1.1x
DT1K	$Q_{a\&d}$	Sib.	11	4.2x	4.4x	18	19x	19x	1.4	1.1x	1.0x
DT1K	$Q_{a\&d}$	Dist	11	4.1x	4.2x	20	21x	21x	1.6	1.2x	1.2x
DT10K	$Q_{a\&d}$	RL	19	12x	12x	22	24x	21x	1.6	1.4x	1.5x
DT10K	$Q_{a\&d}$	DPL	18	12x	11x	20	21x	22x	1.3	1.2x	1.2x
DT10K	$Q_{a\&d}$	Sib.	21	1.3x	1.3x	20	23x	20x	1.4	1.1x	1.1x
DT10K	$Q_{a\&d}$	Dist	21	1.3x	1.3x	20	21x	21x	1.6	1.3x	1.3x
DT100K	$Q_{a\&d}$	RL	80	49x	50x	20	19x	19x	1.1	1.1x	1.3x
DT100K	$Q_{a\&d}$	DPL	81	60x	60x	21	21x	23x	0.85	1.0x	0.75x
DT100K	$Q_{a\&d}$	Sib.	130	0.82x	0.84x	20	21x	21x	1.2	0.94x	0.92x
DT100K	$Q_{a\&d}$	Dist	130	0.80x	0.83x	21	25x	23x	1.4	1.1x	1.1x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
WT10	$Q_{a\&d}$	RL	10	7.5x	7.8x	18	20x	21x	1.8	1.6x	1.6x
WT10	$Q_{a\&d}$	DPL	10	7.5x	7.6x	19	22x	20x	1.3	1.1x	1.1x
WT10	$Q_{a\&d}$	Sib.	9.4	9.1x	9.6x	19	21x	20x	1.4	1.0x	1.0x
WT10	$Q_{a\&d}$	Dist	9.7	9.8x	10x	18	19x	19x	1.5	1.1x	1.2x
WT100	$Q_{a\&d}$	RL	11	6.7x	7.1x	19	21x	20x	1.4	1.4x	1.3x
WT100	$Q_{a\&d}$	DPL	11	7.5x	7.6x	18	20x	19x	1.2	1.1x	0.98x
WT100	$Q_{a\&d}$	Sib.	9.8	8.3x	7.7x	18	19x	19x	1.3	1.1x	1.2x
WT100	$Q_{a\&d}$	Dist	9.8	8.6x	8.8x	21	23x	22x	1.7	1.2x	1.2x
WT1K	$Q_{a\&d}$	RL	13	8.1x	8.3x	21	25x	23x	1.1	1.3x	0.94x
WT1K	$Q_{a\&d}$	DPL	12	7.8x	8.0x	19	22x	21x	1.3	1.0x	0.99x
WT1K	$Q_{a\&d}$	Sib.	11	4.2x	4.3x	21	24x	19x	1.4	1.1x	1.0x
WT1K	$Q_{a\&d}$	Dist	11	4.1x	4.2x	23	24x	25x	1.6	1.1x	1.2x
WT10K	$Q_{a\&d}$	RL	19	11x	11x	21	25x	25x	1.4	1.2x	1.5x
WT10K	$Q_{a\&d}$	DPL	18	11x	11x	21	22x	23x	1.2	1.1x	1.1x
WT10K	$Q_{a\&d}$	Sib.	21	1.2x	1.3x	19	22x	23x	1.3	1.1x	1.1x
WT10K	$Q_{a\&d}$	Dist	21	1.3x	1.3x	21	25x	25x	1.4	1.2x	1.1x
WT100K	$Q_{a\&d}$	RL	80	42x	44x	20	22x	23x	1.6	1.2x	1.2x
WT100K	$Q_{a\&d}$	DPL	81	44x	45x	23	27x	25x	1.3	1.0x	0.96x
WT100K	$Q_{a\&d}$	Sib.	130	0.78x	0.80x	20	23x	22x	1.2	0.96x	0.89x
WT100K	$Q_{a\&d}$	Dist	130	0.79x	0.79x	20	22x	21x	1.4	1.0x	0.98x
SNB/C	$Q_{a\&d}$	RL	15K	7.3Kx	7.9Kx	22	22x	24x	1.6	1.3x	1.4x
SNB/C	$Q_{a\&d}$	DPL	15K	6.9Kx	8.0Kx	19	15x	19x	1.2	1.1x	1.0x
SNB/C	$Q_{a\&d}$	Sib.	16K	4.4x	4.5x	19	18x	22x	1.2	1.0x	1.0x
SNB/C	$Q_{a\&d}$	Dist	16K	4.2x	4.3x	20	19x	24x	1.2	1.1x	1.1x
SNB/P	$Q_{a\&d}$	RL	15K	13Kx	9.7Kx	20	24x	22x	1.4	1.3x	1.3x
SNB/P	$Q_{a\&d}$	DPL	14K	9.2Kx	9.7Kx	17	20x	19x	1.3	1.2x	1.1x
SNB/P	$Q_{a\&d}$	Sib.	14K	4.5Kx	4.6Kx	21	24x	23x	1.2	1.1x	1.0x
SNB/P	$Q_{a\&d}$	Dist	14K	4.5Kx	4.6Kx	18	20x	20x	1.3	1.1x	1.1x
SNB/T	$Q_{a\&d}$	RL	15K	9.0Kx	8.6Kx	19	23x	20x	1.2	1.1x	1.0x
SNB/T	$Q_{a\&d}$	DPL	14K	10Kx	8.6Kx	21	25x	23x	1.4	1.3x	1.3x
SNB/T	$Q_{a\&d}$	Sib.	14K	13Kx	14Kx	19	23x	22x	1.2	1.1x	1.0x
SNB/T	$Q_{a\&d}$	Dist	14K	14Kx	14Kx	18	22x	22x	1.3	0.87x	1.1x

Graph	Qu.	Param.	Apache AGE			Kuzu			Neo4j		
			B	S_D	S_P	B	S_D	S_P	B	S_D	S_P
SNB	IC12	–	190K	1.1x	1.1x	400	1.3x	0.98x	160	1.3x	1.5x
SNB	IS2	–	61K	16x	4.7x	44	0.18x	0.16x	4.5	0.99x	0.42x
SNB	IS6	–	17K	8.0x	5.3x	29	1.3x	0.24x	1.9	0.95x	1.0x